

Sprint Review and Retrospective

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The SNHU Travel project provided a useful opportunity to see how a Scrum-Agile approach changes how software work is planned, completed, reviewed, and improved.

ChadaTech had historically used the waterfall model, but the SNHU Travel pilot required the team to work through iterative development and Scrum events. Instead of treating requirements, development, testing, and delivery as isolated stages that happened once, the project divided work into smaller pieces that could be reviewed and adjusted. The experience demonstrated that Agile is especially valuable when requirements can change and when feedback is needed throughout development.

Applying Scrum-Agile Roles

Each Scrum-Agile role contributed to the success of the SNHU Travel project. The Product Owner focused on the customer and product priorities. This accountability mattered because the team needed to understand what the travel application should accomplish for users, not just complete technical tasks. User stories translated customer needs into manageable work and helped the team decide which features to address first.

The Scrum Master supported the process by helping the team follow Scrum practices, identifying impediments, and encouraging communication. The Scrum Master role was particularly important when the team had to respond to changing requirements. Rather than allowing a change to derail the project, the team could discuss the new priority, adjust the backlog, and determine what could realistically be completed during the next Sprint.

The development and testing responsibilities contributed directly to product quality. Development work turned user stories into working functionality, while testing provided an

independent check that the functionality behaved as expected. My experience with the tester role reinforced the importance of protecting quality while still maintaining an Agile pace. A fast development cycle is not successful if defects accumulate or if the completed feature does not meet the user's needs. Scrum's emphasis on inspection and adaptation supported this balance.

Completing User Stories

The Scrum approach helped user stories reach completion by breaking the overall SNHU Travel application into smaller, understandable increments. A large requirement can be difficult to manage when treated as a single assignment. In Scrum, the team could identify a specific user need, clarify its acceptance criteria, estimate the work, and place it into a Sprint. The team could then develop and test the story before considering it complete.

This incremental approach also made progress easier to see. Instead of waiting until the end of the project to discover that a major feature was incomplete, the team could review working portions of the application during Sprint Reviews. The Definition of Done provided a useful quality standard because a story was not simply complete when coding stopped; it needed to satisfy the team's completion expectations. This reflects the Scrum principle that working product increments provide meaningful evidence of progress.

Handling Interruptions and Changes in Direction

One of the clearest advantages of Scrum during the SNHU Travel project was the ability to respond when priorities changed. In a waterfall process, a significant change after requirements or design had been approved could require revisiting completed documentation, redesigning work, and moving through several dependent stages again. Scrum instead provided a controlled way to incorporate change through the Product Backlog and Sprint planning process.

When the project direction changed, the team could evaluate the new requirement, prioritize it against existing work, and decide how it should affect upcoming Sprint work. This did not mean that every interruption could be inserted immediately without consequences. Instead, Scrum gave the team a framework for making the tradeoff visible. The ability to inspect progress and adapt the plan helped the project continue moving forward without treating the original plan as unchangeable.

Communication and Collaboration

Communication was one of the most important factors in making the Agile process work. Effective communication was specific, timely, and focused on the work rather than on assigning blame. For example, a useful testing message would be: "The travel search feature is returning results, but the filtering behavior does not match the acceptance criteria. I recommend confirming the expected filter behavior before the story is marked Done." This message identifies the problem, connects it to a project requirement, and proposes a next step.

Another effective example would be a status update such as: "The current Sprint goal is still achievable, but the new request will require us to move one lower-priority story back to the Product Backlog." This type of communication gives the team enough information to decide without creating unnecessary confusion. The examples were effective because they encouraged collaboration and made the status of the work transparent. Agile principles emphasize communication and collaboration, and the Scrum events provided regular opportunities to practice them.

Organizational Tools and Scrum Events

The Product Backlog, Sprint Backlog, task tracking, user stories, and acceptance criteria were valuable organizational tools. The Product Backlog provided a prioritized view of the work. The Sprint Backlog narrowed that focus to the work selected for the current Sprint. Tracking individual tasks made progress and impediments more visible, while user stories kept the team focused on user value rather than technical activity alone.

The Scrum events gave those tools a practical purpose. Sprint Planning connected the backlog to a specific Sprint goal. Daily Scrum provided a short opportunity to inspect progress and identify impediments. The Sprint Review allowed the team and stakeholders to examine the increment and discuss whether the product was moving in the right direction. The Sprint Retrospective focused on the process itself, allowing the team to identify what worked, what did not work, and what should change in the next Sprint. Together, these events created a cycle of planning, execution, inspection, and adaptation.

Evaluating the Agile Process

The SNHU Travel project showed several advantages of Scrum-Agile. First, it supported changing requirements without rebuilding the entire project plan. Second, it encouraged frequent feedback and made incomplete or defective work visible earlier. Third, the defined accountabilities and Scrum events created opportunities for collaboration. Finally, iterative delivery reduced the risk of waiting until the end of the project to discover that the product did not satisfy users.

There were also disadvantages. Scrum requires consistent participation and communication, and those activities require time. Frequent meetings can become inefficient if they lack focus. Agile also depends heavily on the team's ability to prioritize work and make

decisions. If the Product Backlog is poorly managed or requirements are constantly changed without considering capacity, the team can lose focus. In addition, moving from waterfall to Scrum can require a cultural change because team members must become comfortable with transparency, shared responsibility, and adaptation.

For the SNHU Travel project, Scrum-Agile was the better approach. The application was a customer-facing software product, and the scenario included changing needs and the possibility of discovering better requirements during development. Those conditions favored iterative development and frequent feedback. A waterfall approach could still be appropriate for projects with highly stable requirements, extensive regulatory documentation, or limited expected change. However, the SNHU Travel project benefited from the flexibility, inspection, and incremental delivery Scrum provided.

Conclusion

The most important lesson from the SNHU Travel project is that Agile is not simply a collection of meetings or a faster version of waterfall. Its value comes from creating a repeatable cycle in which a team establishes a goal, completes a useful increment, receives feedback, and adapts. The roles, user stories, communication practices, organizational tools, and Scrum events worked together to make that cycle possible. Based on the project experience, ChadaTech should consider expanding Scrum beyond the pilot team, while recognizing that successful adoption would require training, clear accountability, effective communication, and a willingness to improve the process over time.

References

Agile Manifesto. (2001). Principles behind the Agile Manifesto.

<https://agilemanifesto.org/principles.html>

Ågren, P., Knoph, E., & Berntsson Svensson, R. (2022). Agile software development one year into the COVID-19 pandemic. *Empirical Software Engineering*, 27, 121.

<https://doi.org/10.1007/s10664-022-10176-9>

Mishra, A., & Alzoubi, Y. I. (2023). Structured software development versus agile software development: A comparative analysis. *International Journal of System Assurance Engineering and Management*, 14, 1504–1522. <https://doi.org/10.1007/s13198-023-01958-5>

Verwijns, C., & Russo, D. (2023). A theory of Scrum team effectiveness. *ACM Transactions on Software Engineering and Methodology*, 32(3), Article 74. <https://doi.org/10.1145/3571849>

Schwaber, K., & Sutherland, J. (2020). The Scrum Guide: The definitive guide to Scrum: The rules of the game. Scrum.org. <https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>